Design and Finite Element Analysis of IoT based Smart Helmet

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Abstract*— As mentioned in the global two-wheeler market report, the two-wheeler market is projected to surpass $ 125 billion by 2024. According to WHO statistics, approximately 1.35 million people die each year as a result of road traffic crashes. And more than half of those are pedestrians, cyclists, and motorcyclists. Correct helmet use can lead to a 42 percent reduction in risk of fatal injuries and a 69 percent reduction in the risk of head injuries. These are concerning statistics. These growing numbers call forth new and upcoming technological advancements concerning the safety of the users. The main reason behind coming up with this smart helmet is the very issue of safety. The main objective of the project is to build a smart system which is integrated with sensors and modules for the safety of the user. While working on improving the safety of riders, it was ensured that its aesthetics and ergonomics are not compromised.

***Keywords*— IoT, FEA, Smart Helmet**

I. Introduction

Advancements in motorcycles have been faster than in helmets. Irrespective of whether one rides a road bike, mountain bike, or motorcycle, a good helmet is a must to stay safe. But helmets can turn uncomfortable after a sustained period of use. And navigating with the help of maps while having a helmet on is not very convenient. So basic concepts of IoT are used in the design.

A Bluetooth module enables connectivity to the smartphone. To ensure rider safety, vibration sensors detect collision and location information can be communicated. Other features also ensure rider safety. We had proposed a basic idea of our smart helmet which incorporated the user's safety as well as the user's needs.

For user safety, the impact sensor interfaced with the GPS module has been used to detect if the user comes across any accident. Texas Instruments kits are used for noise cancellation which is able to provide noiseless transmission between the users using Bluetooth module also providing the consumer a breath-taking surround-sound experience. The basic components of the helmet are Shell (Outer Layer of helmet), EPS Liner (High density foam to absorb shocks), Comfort Liner (Ensures comfort of head), Cheek Pads (for protection from side impacts), Visor (movable front shields to protect the face of the users), and a Smart Buckle system.

II. Literature Review

The goal of this literature survey is to compare the several approaches and methodologies that have been published in the field of smart helmet design.

Smart Helmet for Safe Driving: The use of FSR to detect whether helmet is worn by the user, MQ-3 alcohol sensor to detect whether the driver is drunk or not, vibration sensors to detect vibration and RF module for communication are the modules proposed in the work for smart helmet in [2].

A smart helmet using GSM and GPS with module SIM808, a limit switch to detect whether helmet is worn or not and accelerometer for continuous speed checking is proposed in a further work [3].

In reference [14], they have used FSR to check whether helmet is worn or not and accordingly the bike will start, a camera is placed in the bike which will check for the helmet. They have used MQ3 i.e. alcohol sensor to check the

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permissible range of alcohol to start the bike and message will be sent to registered number. GPS system in bike will track the location and send message if injured

The research paper [4] they used a limit switch for checking of helmet worn by the user or not, accelerometer sensor for keeping track of speed and if exceeding the ignition gets cut, GSM module to communicate the message. [5] provides the basic insights regarding the oblique impact of the motorcycle helmet and the angular acceleration of the head foam due to various factors.

[6] Provides vital information about the conventional design of the helmet and the benefits of using helmets along with a statistical case study of accidents in Thailand.

Research paper [11] gives the basic understanding of the safety limits for impact testing for helmets as per the BIS (Bureau of Indian Standards). In the study conducted in [13], a car accident was reconstructed in order to simulate and analyze the motion, impact velocity, impact angle in the collision.

Based on the current evidence, using smart helmets helps in reducing injuries as numerous studies are introduced by different authors all over the world and these innovations will prove beneficial for the people.

III. Market Analysis in India

According to a sample survey conducted by us in Pune, India, 25% of the people preferred cabs for daily commuting. This was linked to the Ola, Uber boom in India. 15% people used 4-wheelers for commute. 10% prefer public transport and the rest use mopeds and bikes. About 50% of people that we surveyed used 2- wheelers, which was due to the fact that 2-wheelers are cheaper, compact, and easily maneuverable. Due to the decisions of the Government of India to improve and focus on road safety, it is mandatory for 2-wheeler users to wear helmets, failing which can attract hefty fines. In spite of the introduction of these strict traffic regulations, 5% of drivers did not wear helmets. Many also felt the need for an inbuilt navigation system because using smartphones for navigation can sometimes lead to accidents and isn't that reliable. They also complained regarding the bulky structures of the helmets and acute neck and back pain. The need for an ergonomic helmet was felt. Another big problem is about the post-accident situation. Very few people readily come forward to help an injured biker in India which is the harsh reality. According to another national survey, about 37 million 2-wheelers run on the roads in India. Hence, the necessity of incorporating an accident detection and communication system in helmets is crucial.

IV. Working of the Circuit

The block diagram explains the basic functions of the various modules of the Smart Helmet system. As safety is of paramount importance in our product, it is designed in such a way that the helmet's features would not start up until the rider is buckled in. The capacitive sensor is placed inside the buckle of the helmet. So only when the user ties the strap and locks the buckle of the helmet will the sensor get actuated. This idea ensures that the users take the basic safety.

The SW-420 sensor will be continuously sensing and if the user meets with an accident, that is vibrations frequency is greater than the preset threshold frequency (40 Hz), then the controller board will become high and the signal will be sent to the GSM module. Fig. 01 shows the block diagram of the system.

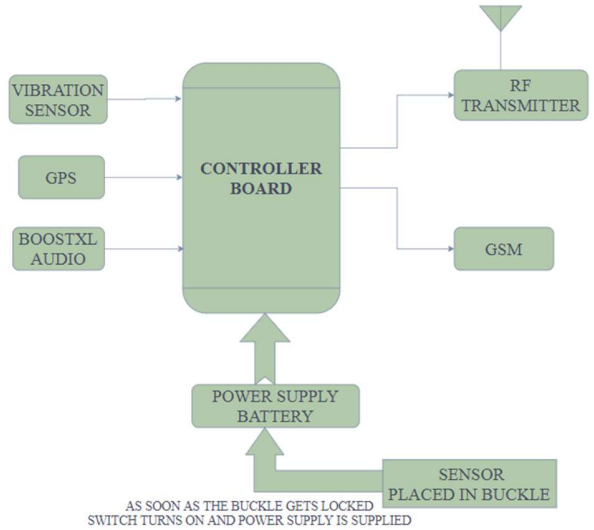


Fig. 01 - Smart Buckle System Block Diagram

A message of the user's location with the help of GPS module will be sent to the desired port via use of the GSM module. The port will receive this message in the form of an NMEA string. The flowchart is shown in the Fig. 02.

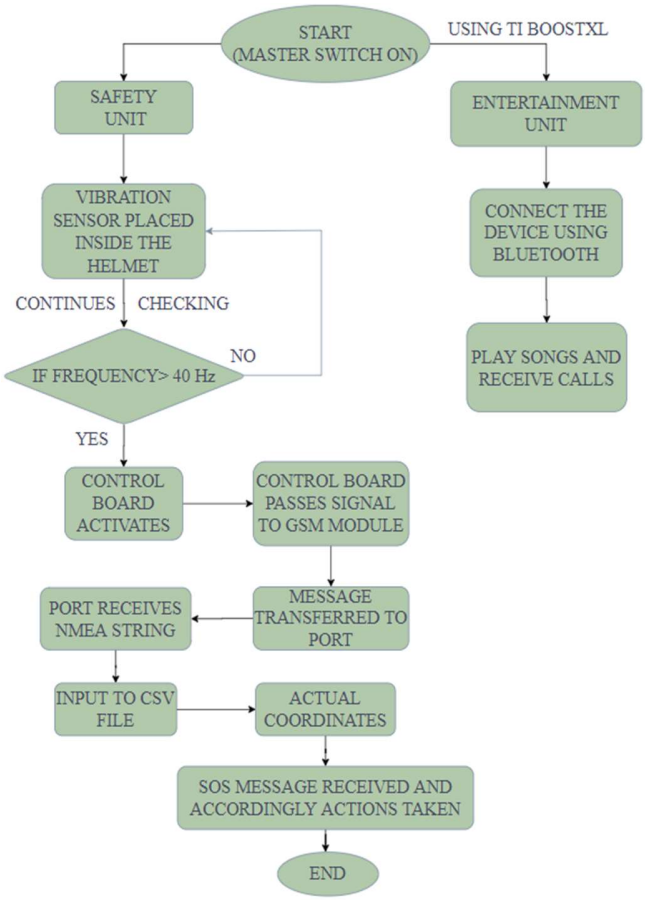


Fig. 02 – Flowchart of the System

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V. Implementation Setup

A. Components Required

1) SW-420 Vibration Sensor
2) NEO-6MV2 GPS Module
3) Controller Board
4) SGW 8130 BT sensor tag
5) GSM Module (SIM900A GSM Modem)
6) TI BoostXL

B. SW-420 Vibration Sensor

The vibration sensor is shown in the Fig. 03

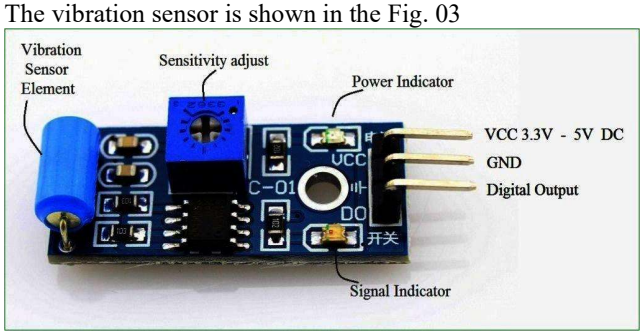


Fig. 03 – SW-420 Vibration Sensor

Vibration Sensor SW-420 is a highly sensitive non-directional sensor used to sense the vibrations which are placed inside the helmet. The position of vibration sensors is selected to be at the back side and are placed in such a way that the accuracy as well as the efficiency would be maximum. Placement is done such that when placed it would form an equilateral triangle with side length at 50mm all over the helmet. The sensor is interfaced with the controller board and GSM module and the threshold frequency of vibrations set to 40 Hz. The sensor would be continuously sensing the vibrations, it would be working in IF ELSE condition i.e. if the sensor senses frequency greater than the desired or set frequency a message would be sent via GSM to the nearest port and necessary actions would be taken. And in other conditions it would recheck and sense continuously. The size of this module is 30mm x 15mm.

C. NEO-6VM2 GPS Module

Fig. 04 shows the actual GPS module.

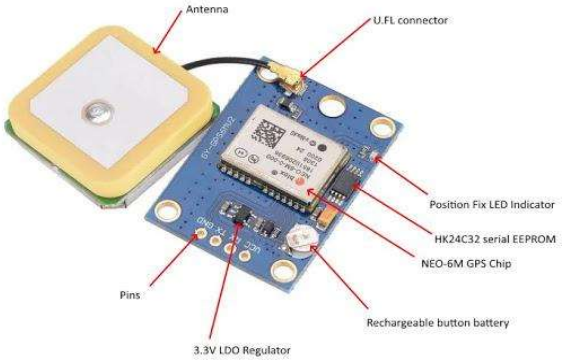


Fig. 04 – NEO-6MV2 GPS Module

The NEO-6MV2 is a GPS (Global Positioning System) module and is used for navigation. The module simply checks its location on earth and provides output data which is longitude and latitude of its position. The NEO-6VM2 GPS Module is interfaced with a controller board occupying two pins. Vibration sensors would perform their task and if a message is to be delivered then only this module would come into action. The interfacing between the GPS module and the controller board is performed by using a +5V supply from the power side to the controller board and any ground pin. Any two pins would work for the serial communication. The GPS receiver module gives output in standard NMEA i.e. National Marine Electronics Association string format. It provides output serially on a Tx pin with default 9600 baud rate. This NMEA string output from a GPS receiver contains different parameters separated by commas like longitude, latitude, altitude, time etc.

D. SGW 8130 BLE sensor tag

The SGW 8130 sensor tag is a highly integrated Bluetooth Low Energy (BLE) device with different sensor options. It reports temperature, humidity, light intensity and motion data through a wireless Bluetooth 4.0 link. Bluetooth sensor tag is interfaced with the controller board. The Booster audio module provided by TI is also interfaced with the Bluetooth module. It would be an additional feature included in our product. Music will be played when required by the user and other times it will guide the user with directions to the required location. Another feature of this product is for biker gangs which would help them for a faster and easier way of communication.

E. SIM900A GSM Modem

Fig. 05 represents the GSM Module.



Fig. 05 – SIM900A GSM Module

The SIM900A is a complete Dual-band GSM/GPRS solution in a SMT module which can be embedded in the customer applications. Featuring an industry-standard interface, the SIM900A delivers GSM/GPRS 900/1800MHz performance for voice, SMS, Data, and Fax in a small form factor and with low power consumption. GSM digitizes and compresses data, then sends it down a channel with two other streams of user data, each in its own time slot. GSM is used for transferring the SOS message.

F. TI BoostXL

Texas Instruments BoostXL is a high-quality audio playback with onboard 14-bit DAC (can be bypassed if target MCU has an integrated DAC). It can automatically switch from onboard speakers and microphone to a headset with microphone where

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onboard microphone supports sampling rates of up to 20kHz.This module is used as a part of our Entertainment Unit. Fig. 06 shows the detailed circuit board of the BoostXL.

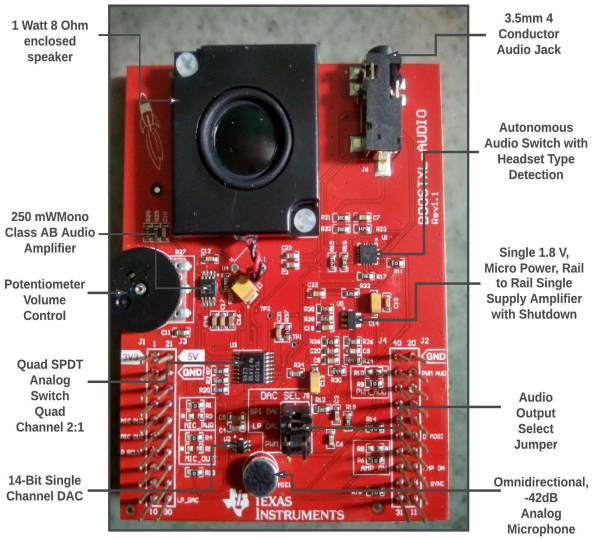


Fig. 06 – TI BoostXL

When plugged into a LaunchPad Development Kit, the BOOSTXL-AUDIO Audio Booster Pack Plug-in Module adds audio input functionality from a microphone, as well as audio output through an on-board speaker. Headphone input and output is also supported, and automatically enabled when a plug is inserted into the Booster Pack. This audio I/O stream lets the user experiment with the digital signal processing (DSP) and filtering capabilities of the microcontroller found on the attached LaunchPad Development Kit.

VI. Circuit Diagram

Fig. 07 shows the circuit diagram of the safety unit.

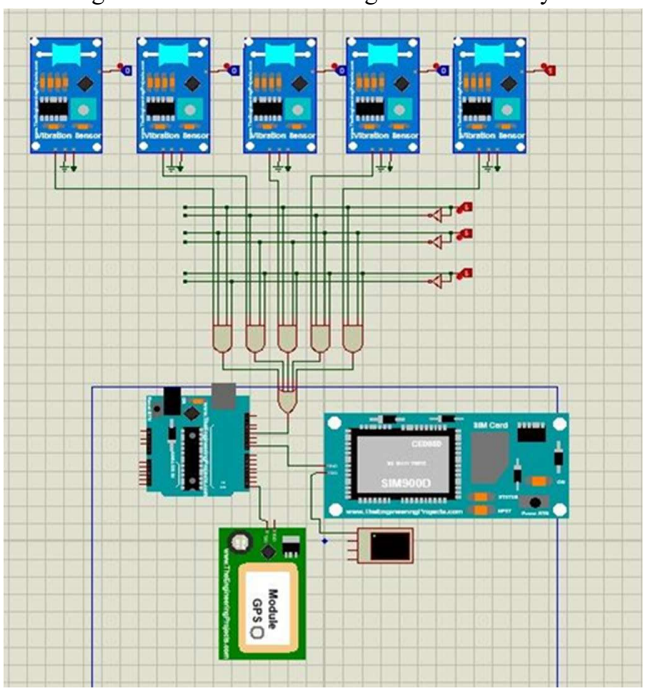


Fig. 07 – Circuit Diagram of the Safety Unit

VII. Construction of the Helmet

This capacitive sensor would be the most important core component of the system as it would act as an input to the power supply, because of which the entire system would start. So, if the rider simply puts on the helmet without locking, he/she will not be able to access the various features other than the basic impact protection. This idea ensures that the users take the basic safety precautions prior to driving, thus instilling in their minds the necessity of safety on roads.

After the buckle is locked, the Safety Unit and Entertainment Unit activate. The smartphone can be paired with the SGW 8130 BLE sensor tag and the rider can enjoy music and receive calls. The TI BoostXL module provides excellent sound quality for this purpose. In the Safety Unit, the SW-420 Vibration sensors continuously sense the external vibrations. They are preset at 40Hz of threshold vibration frequency. If the rider meets with an accident, vibrations of frequency more than 40Hz are produced during the impact and the sensors activate the control board. The control board then passes a signal to the GSM module and the GPS module. The GPS module detects the location and sends an NMEA string to the control board. This NMEA string serves as an input to the CSV file conversion algorithm written in Python language and the global coordinates of the current location are obtained. This data is then broadcasted in the form of an emergency SMS through the GSM module to the port designed so that necessary actions are taken.

VIII. Module Placement

The sensor placement is done according to the space available between the outer shell and the inner foam layer. The placement is designed based on the individual sensor sizes. The Fig. 08 shows the module placement briefly inside the helmet.

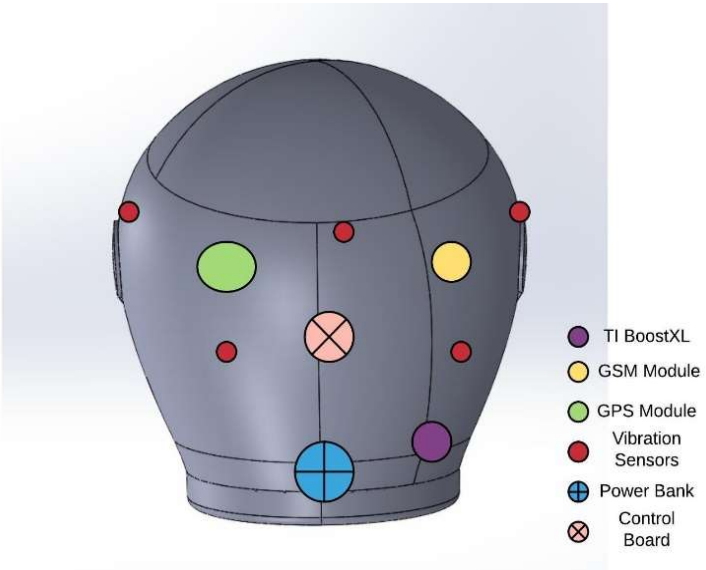


Fig. 08 - Sensor and Module Placement (Rear View)

Here the sensors and modules are placed in the rear part of the helmet. These modules are embedded in a foam casing to safeguard them against vibrations and other foreign entities. Fig. 09 represents the flowchart of the code.

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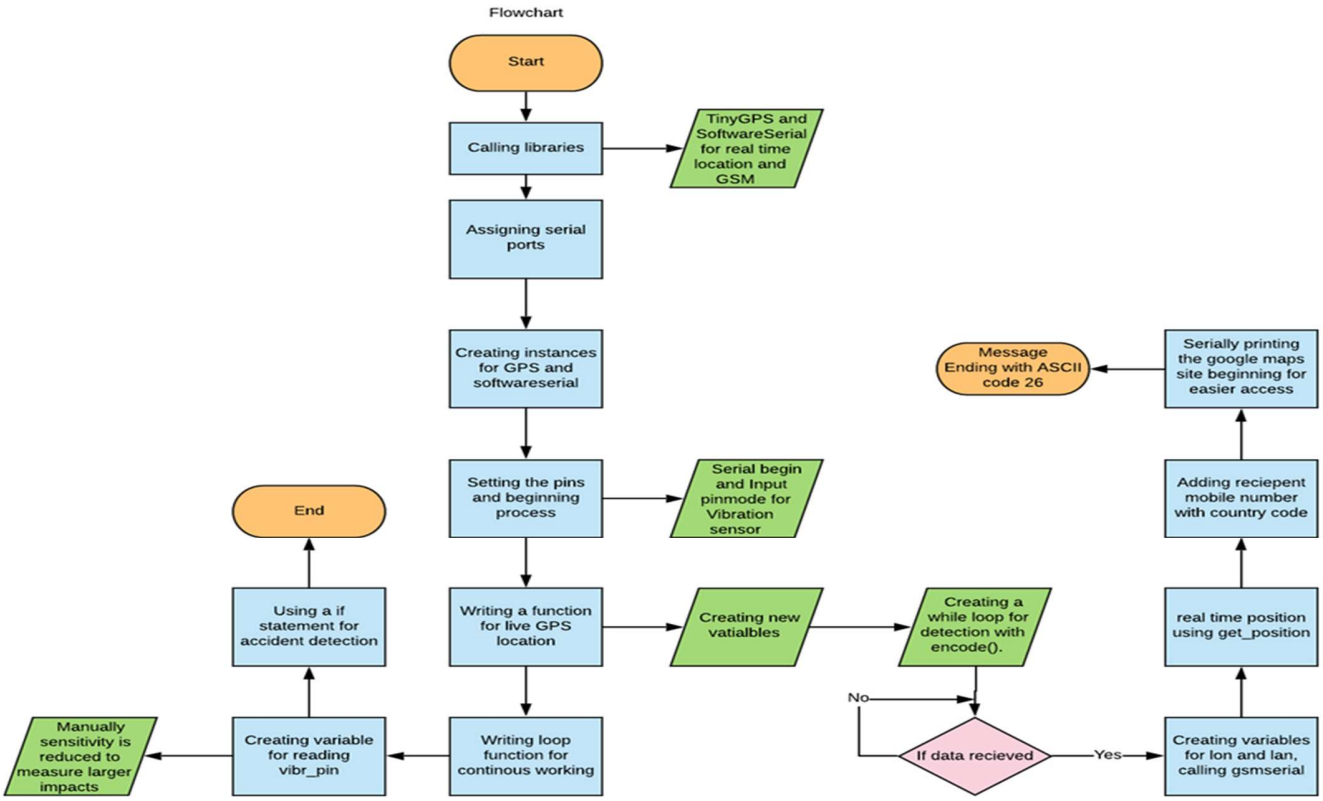


Fig 09. Flowchart of code

IX. Replaceable Battery System

The source of power for the helmet will be replaceable battery packs. 2 battery packs will be provided. The users will be able to recharge one pack after its power dies down and can replace it with the second one. This will provide uninterrupted usage of the helmet, in that the user will not have to wait till the battery recharges and can quickly switch over to the other pack.

X. Advantages

The detection feature is very useful due to which immediate and essential, prompt, and speedy help can be given. The Bluetooth is enabled for music as well as receiving and making calls. The Smart safety buckle ensures that the rider is wearing the buckle at all times as it is essential to hold the helmet in place during an accident. This system does not require any smart-phone app and can be accessed from any device with Bluetooth capability.

The helmet is cheaper than other smart helmets in the market and offers various extra features and additionally, it is considerably light and weighs only 1723 gm including all the electronic components. The smart safety buckle system eliminates the false activation of the system when the helmet accidentally falls from the hands while carrying. The replaceable battery system is easy to use and is practical.

XI. Simulation

The safety system was simulated by applying real-time accident constraints and conditions.

The circuit is simulated on Proteus Software, and the SOS message obtained with the location of the accident in the form of a Google Maps link. The circuit simulation is shown in Fig. 10.

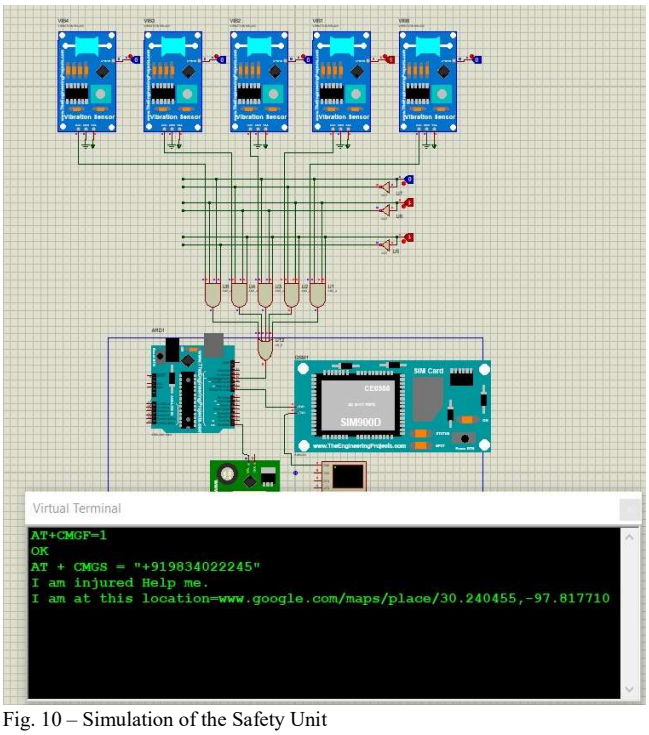


Fig. 10 – Simulation of the Safety Unit

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XII. Design and FEA of Helmet

A. Components Required

The helmet body was designed using SolidWorks 2018 software and the materials were selected based on their strength, weight and durability. Dimensions of the helmet are – 217mm×193mm×202mm (length, breadth and height respectively). Fig. 11 shows the CAD design of the helmet.

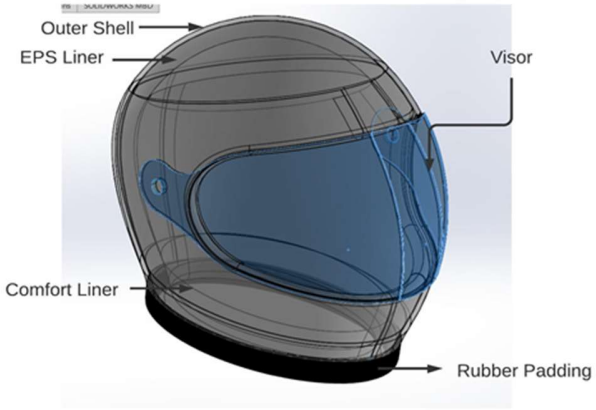


Fig. 11 – CAD design of the helmet

The Fig. 12 represents the helmet part and the materials used. Different layers and parts were designed including the Outer Shell which forms the outermost protective layer of the helmet, EPS Liner or the layer under the outer shell, a comfort liner which makes up the innermost layer that comes in contact with the rider and a rubber padding for perfect fitting.

With all the sensors and modules placed, the ready to use helmet weighs around 1723 gm which is light compared to other normal helmets where the average weight range is around 1600-1800 gm.

TABLE I
MATERIAL PROPERTIES OF HELMET COMPONENTS

Helmet Part	Material	Properties
Outer Shell	Polycarbonate	Young's Modulus = 2.1 GPa Density = 1.22 g/cm^3 High Strength to Weight Ratio
EPS Liner	Extended Polystyrene	Young's Modulus = 3 GPa Density = 0.64 g/cm^3 Ductile, Excellent Damping Properties
Comfort Liner	Polyurethane Foam	17% Spandex, 83% Nylon (40 denier) Breathable, Soft, Lightweight
Cheek Pads	Polyurethane Foam	Aerospacer Mesh Enhances Airflow
Visor	Acrylic	Durable and Excellent Light Distortion Property
Buckle Straps	Nylon	Excellent Strength and Durability
Rubber Padding	Vulcanized Rubber	Excellent Stiffness and Durability

Fig. 12 – Material Properties of Helmet Components

B. FEA of the Helmet

Finite element analysis was performed and the corresponding forces and constraints were applied on the system to simulate an impact situation similar to a real-life accident. A 19.5kN load was applied to the outer shell of the helmet as per the BIS (Bureau of Indian Standards) standards for helmets and the results were recorded. A tetrahedral mesh was generated and the analysis was performed. The Fig. 13 shows the mesh generation.

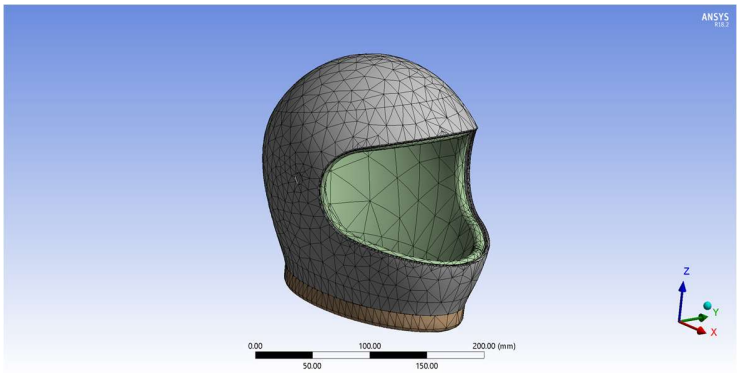


Fig. 13 – Mesh Generation for FEA

The static structural deformation analysis was performed using the ANSYS Workbench 2018.2 software. Fig. 14 shows the static deformation analysis for the top portion loading.

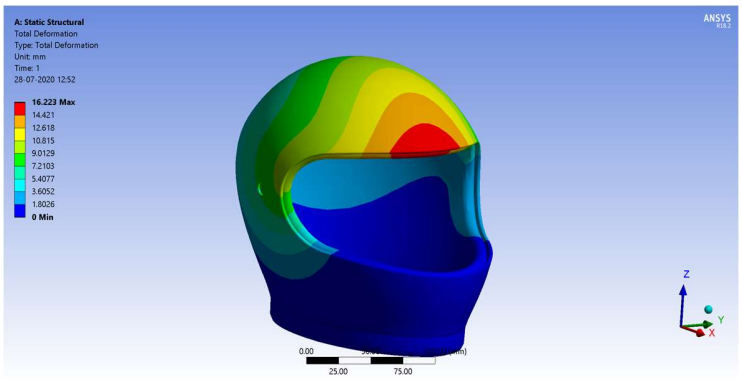


Fig. 14 – Static Deformation Analysis (19.5 kN applied to the top portion)

The maximum deformation was found to be 16.2 mm which is under the acceptable limit of the BIS standard which lies between 6 - 24mm This result showed the robust and strong design of the outer shell of the helmet.

In the second case, 19.5 kN load was applied to the side portion of the helmet body and the results were recorded. The maximum deformation was found to be 9.032 mm which lies in the acceptable limit of the BIS standards. The Fig. 15 shows the static deformation analysis for the side portion loading.

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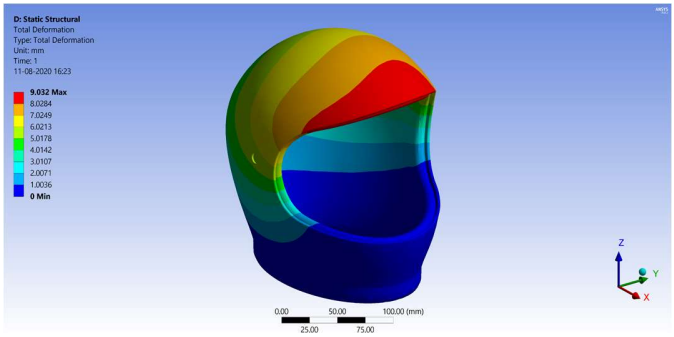


Fig. 15 – Static Deformation Analysis (19.5 kN applied to the side portion of the helmet)

For this iteration, the Equivalent Stress and Strain percentage were found to be 157.28 MPa and 4.766% respectively. Fig 16 shows the equivalent stress analysis. Whereas, the strain analysis is shown in the Fig. 17.

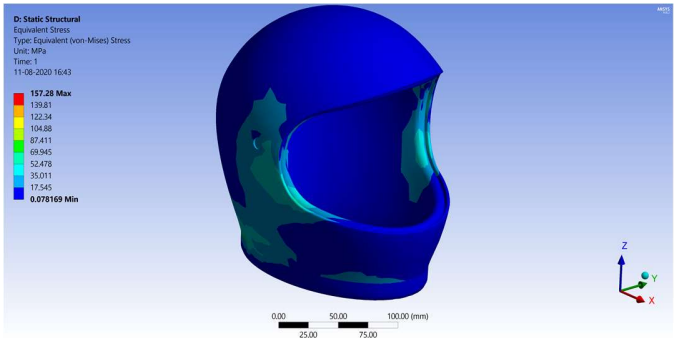


Fig. 16 – Equivalent Stress Analysis

Comparing these results to the study published in [11], the stress and strain values are under the acceptable limit.

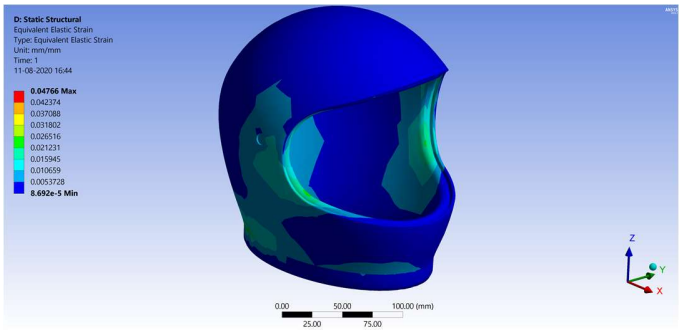


Fig. 17 – Equivalent Strain Analysis

Dynamic analysis of the impact was also performed where the helmet was given a velocity of 10 m/s and the side impact condition was simulated. A concrete block was used for this impact. This analysis was used to predict the real-life situation of an accident and the extent of damage the helmet body would suffer. The impact analysis condition is shown in Fig. 18. In this figure, the yellow body is the entire helmet and the concrete block can be seen resting as a rigid body. The material properties were given to the parts and the analysis was carried out.

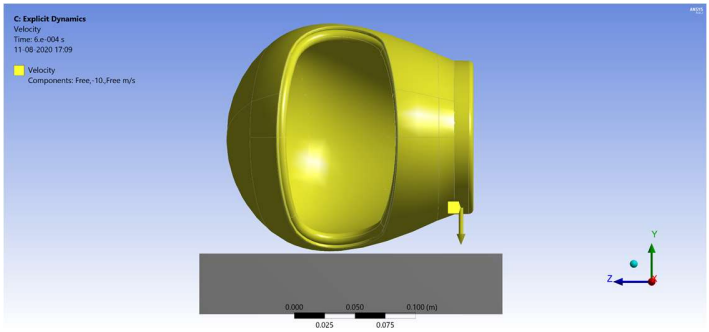


Fig.18 – Impact Analysis Condition

The stress values were calculated and the maximum stress was found to be 2.75e9 Pa and can be seen in Fig. 19.

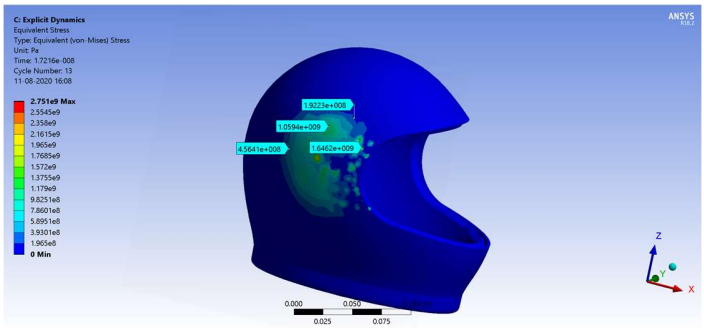


Fig.19 – Dynamic Analysis (Equivalent Stress)

From the above results, it was found that the design of the helmet was safe and the deformation values were under the acceptable limit of the BIS standard for helmet testing.

XIII. Product Comparison

The following comparison table given in Fig.20, compares our idea with five other products already in existence. They have been compared as follows -

TABLE II
PRODUCT COMPARISON WITH EXISTING PRODUCTS

Product	Bluetooth	GPS	Accident Detection	Cost	Weight (gm)
Our Product	YES	YES	YES	$225	1723
Headfox MX	YES	YES	NO	$115	1830
Greenstone G6	YES	YES	NO	$135	2320
Quinn Spitfire Rosso	YES	YES	YES	$400	1750
ILM Bluetooth Integrated Helmet	YES	YES	NO	$192	2000
Harley Davidson Boom! N02	YES	YES	NO	$550	1680

Fig. 20 – Product Comparison with Existing Products

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XIV. Future Scope

This system is designed keeping in mind the road safety norms and convenience. The current system is designed to accept single emergency contact but in the future a provision can be made to store multiple emergency contacts. Further, this system can be programed to send the SOS response to the nearby medical service by incorporating AI and Google cloud services to enhance the emergency response mechanism. With the advent of 5G, a speaking assistant can be embedded into the system itself. Further, an in-built solar powered battery system where the solar panels will be placed on the portion of that helmet that experiences least damage such as the rear portion can also be incorporated to improve the functionality.

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